

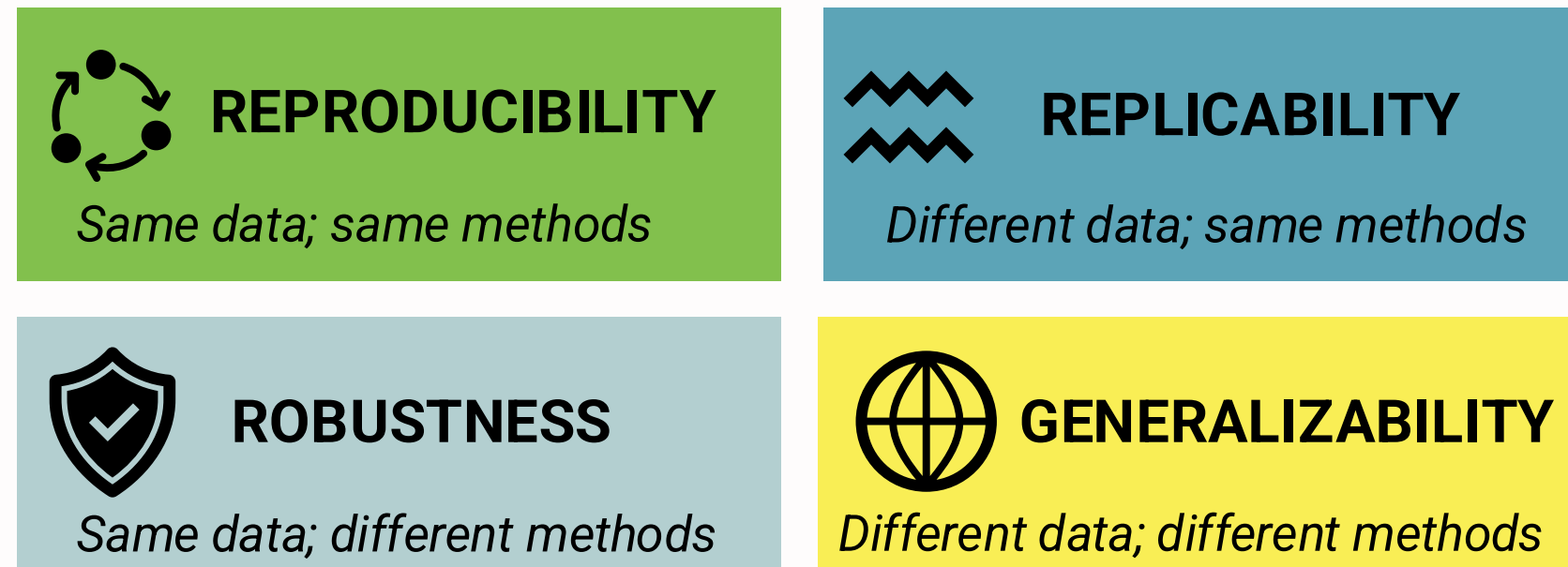
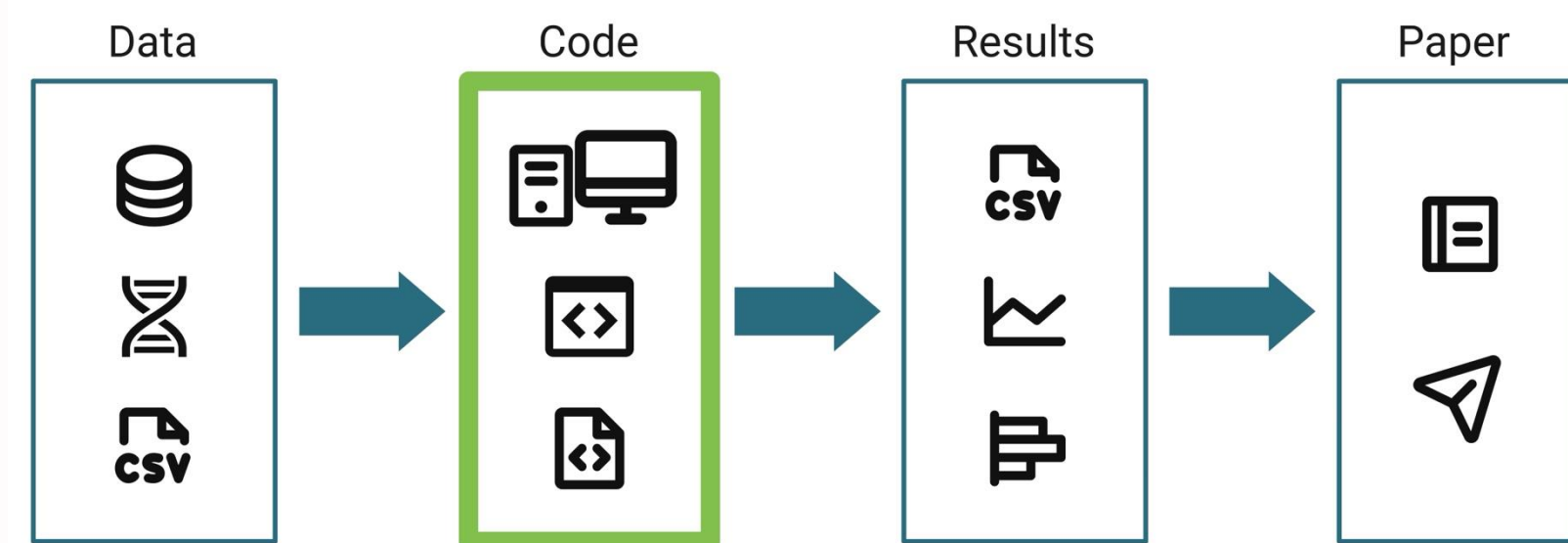
Anatomy Of A Modular Toolchain For Reproducible Bioinformatics Workflows

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INTRODUCTION

Reproducibility is the foundation of trustworthy data-intensive biological research. Our framework implements core reproducibility principles across our software suite.



Schloss PD. 2018. 10.1128/mBio.00525-18.

BEST PRACTICES

Version Control

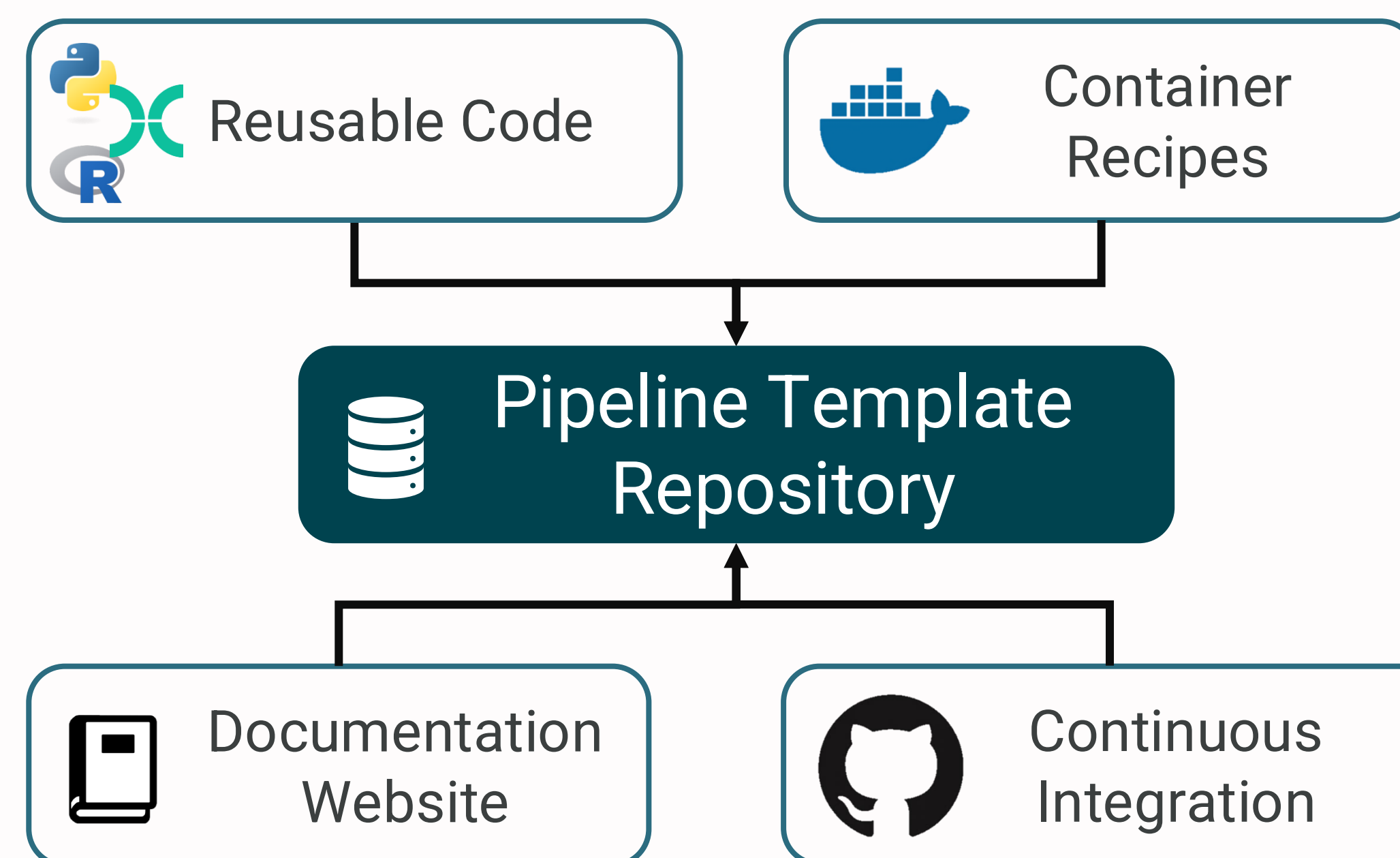
Modular Design

Automated Tests

Documentation

We enforce consistency through templates, modularity, and automation to prevent technical debt, streamline maintenance, and enable long-term sustainability.

TOOLCHAIN FOR REPRODUCIBLE BIOINFORMATICS



Unified framework from template to deployment with automated quality checks.

Fix once;
deploy everywhere

Consistent CLI &
Implementation

Automated testing &
quality checks

PIPELINES BUILT

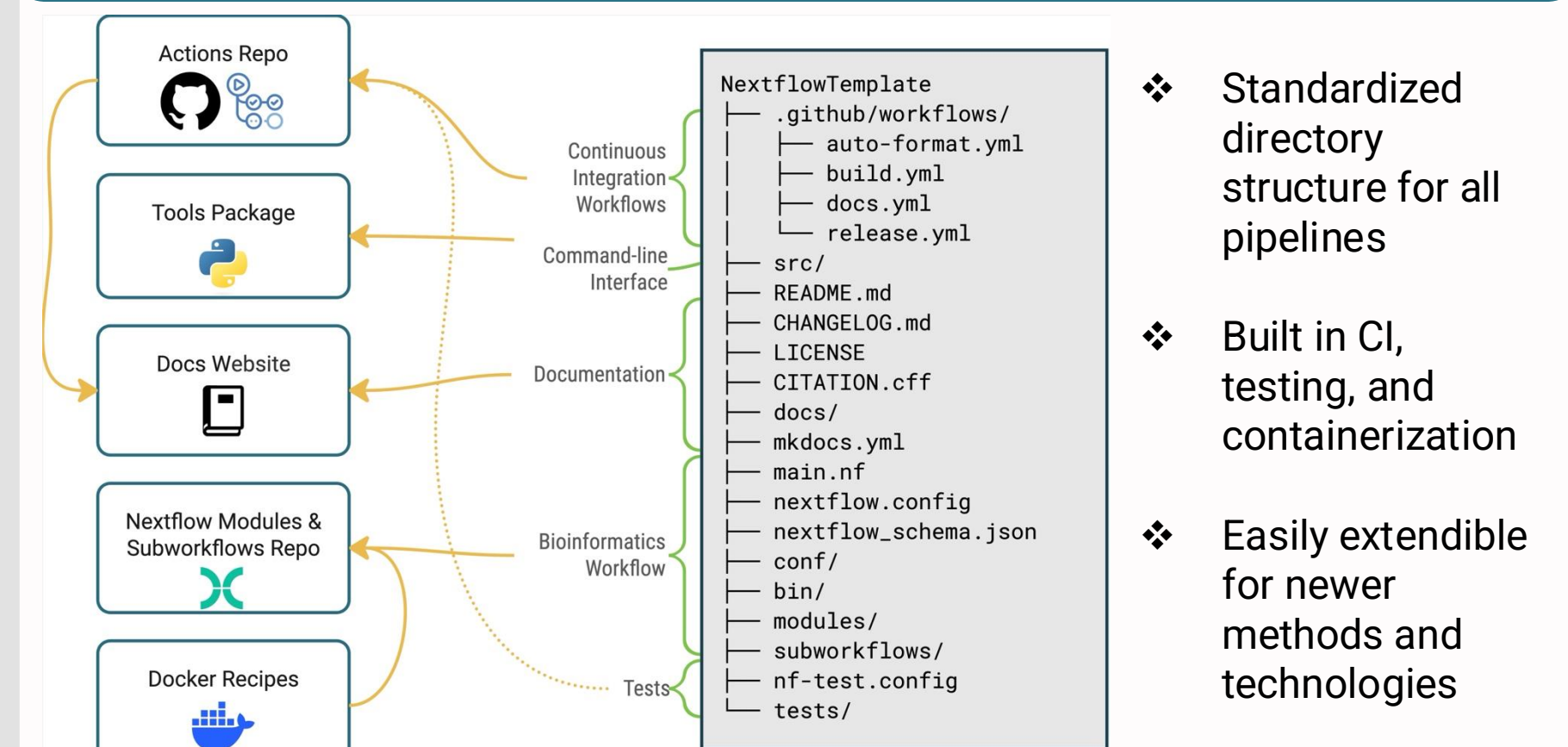


10+

pipelines on NIH Biowulf HPC cluster

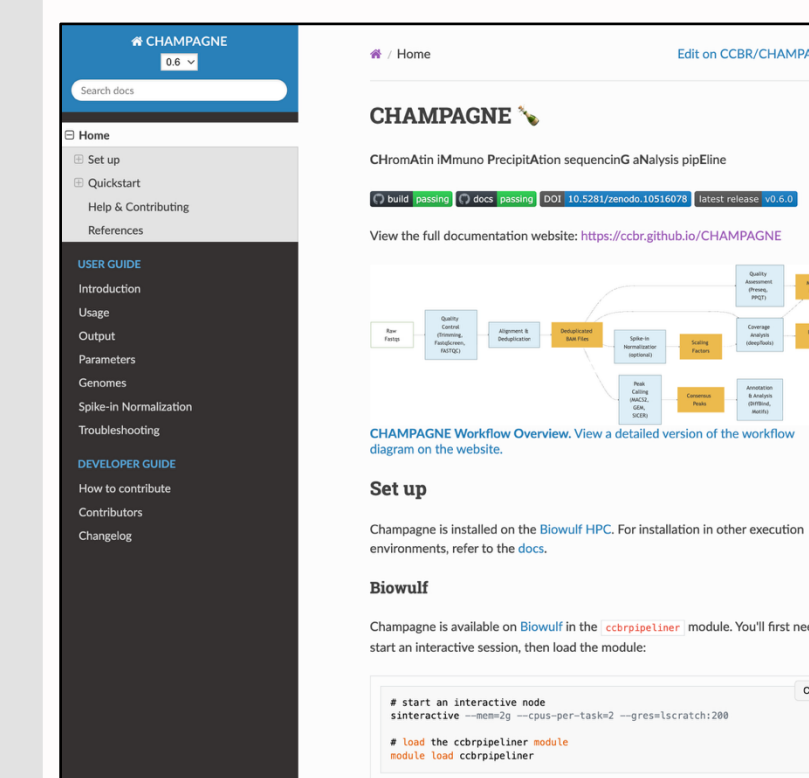
\$ module load ccbripeliner

PIPELINE TEMPLATE



- ❖ Standardized directory structure for all pipelines
- ❖ Built in CI, testing, and containerization
- ❖ Easily extendible for newer methods and technologies

CLI, DOCUMENTATION, AUTOMATION

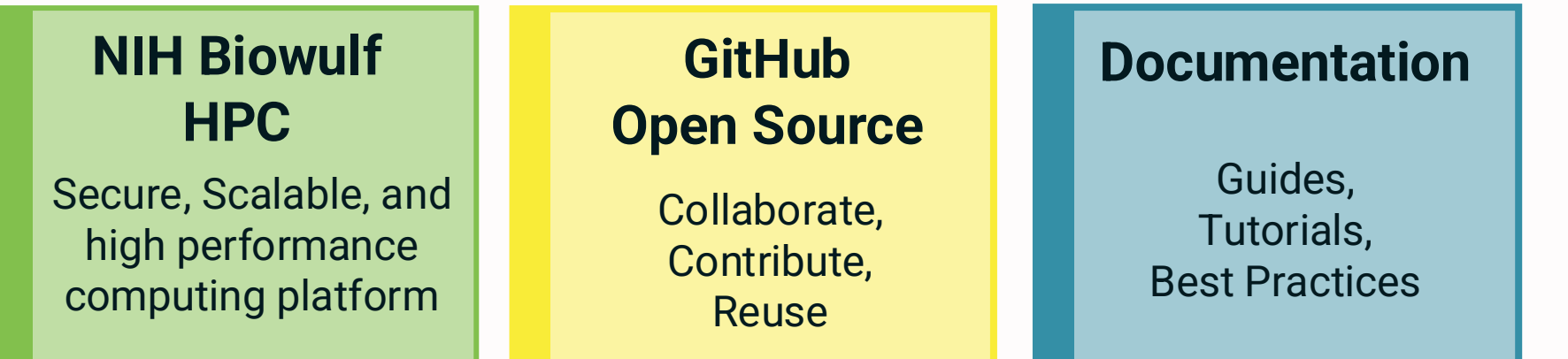


```
$ champagne run --input samplesheet.csv
$ Sinclair run --input samplesheet.csv -params-file ...
```

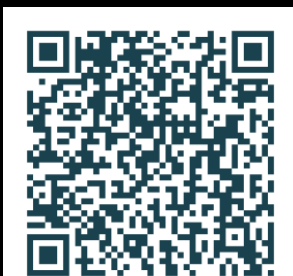
Consistent usage guidelines and documentation across all pipelines simplifies user training.

GitHub Actions runs CI checks on every change.

Scalable, Reproducible, Sustainable Bioinformatics



This Poster



Template Repo



Advancing reproducible computational research through standardized, containerized, and automated bioinformatics pipelines

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